Operating Systems Technologies

University of Minho

Guide 0
Warmup

2026

Goal

This guide aims to set up the lab environment for the Operating Systems Technologies class. We will use VirtualBox to launch a Virtual Machine (VM) with the Ubuntu 25.04 LTS Operating System (OS).

Tools/technologies. We will be working with the VirtualBox hypervisor and we will use Ubuntu 25.04 LTS, Desktop edition, as the OS. The SSH protocol will be used to connect to the VM from the Host computer.

Learning Outcomes. Experiment Linux VM deployment with the VirtualBox hypervisor. Learn how to set up the SSH protocol and keys. Try the Bash environment with different commands.

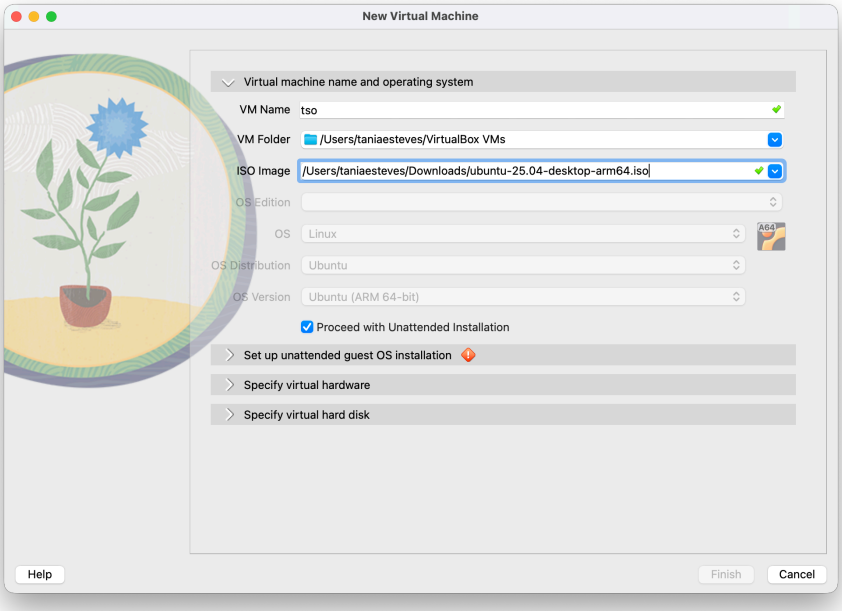
1 Software Setup and Download

1. Install and set up the VirtualBox hypervisor on your host OS (*i.e.*, the operating system running on your laptop):
   (a) Download and install **VirtualBox v7.2.4**: `https://www.virtualbox.org/wiki/Download_Old_Builds_7_2`
   (b) Create a new Host-only network adapter for VirtualBox with the following steps:
      i. Navigate to the Network menu (**File > Tools > Network**) and, on the "*Host-only Networks*" tab, create a new adapter (by clicking the "**Create**" button).
      ii. In Windows/Linux, ensure the Network adapter's IP is set to "192.168.57.1", the DHCP Server's IP is set to "192.168.57.2", "*Lower Bound*" is set to "192.168.57.101", and the "*Upper Bound*" is set to "192.168.57.254". In macOS, adjust the lower and upper bounds to the same values.
2. Download the ISO image of **Ubuntu Desktop 25.04**:
   - For **Intel** and **AMD** processors, download the ISO *"64-bit PC (AMD64) desktop image"* from `https://releases.ubuntu.com/plucky/`.
   - For **ARM** processors, download the ISO *"64-bit ARM (ARMv8/AArch64) desktop image"* from `https://cdimage.ubuntu.com/releases/plucky/release/`.

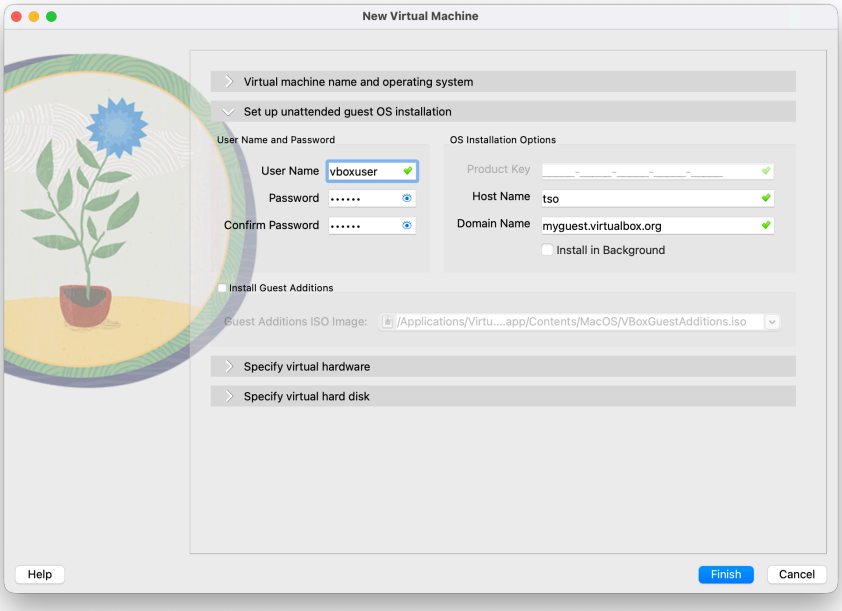
2 Tasks

2.1 Virtual Machine - Install and Configure

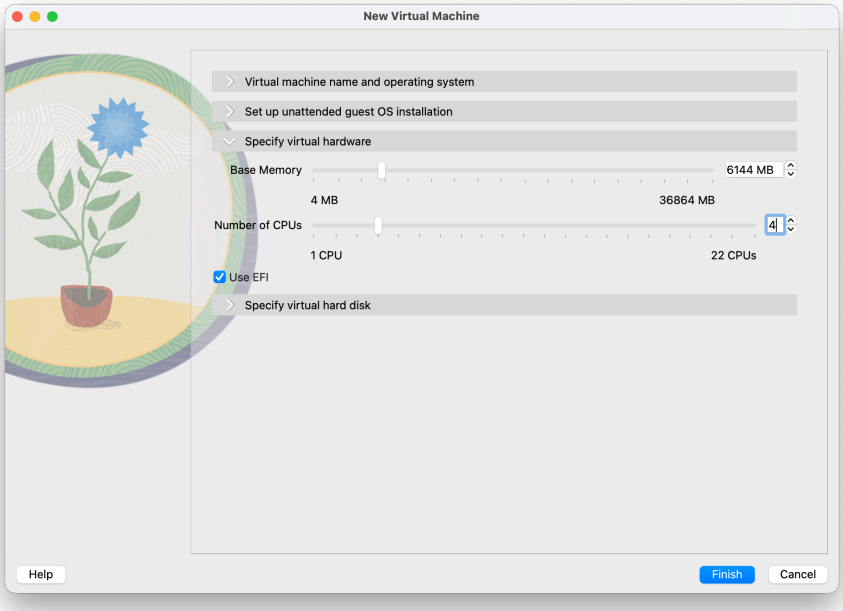
1. Create a new Virtual Machine using the downloaded ISO image.



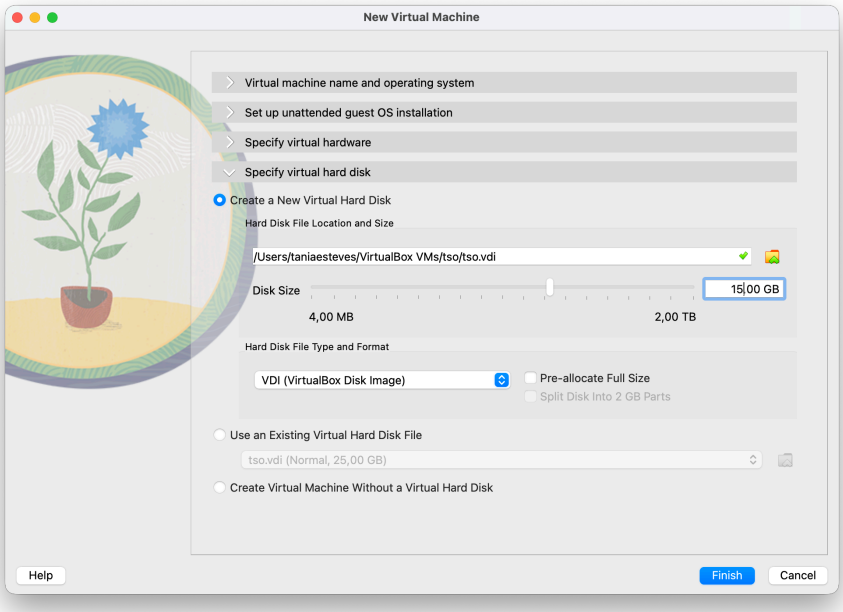
2. Set up your User name and password.



3. Specify the virtual hardware: 6144 MB (6 GB) of RAM and 4 CPUs.



4. Create a new virtual hard disk with 15 GB.



5. Click on *"Finish"* to start creating the new VM.
6. Use the VirtualBox console to connect to the VM. Then, open a terminal and upgrade your OS packages.

```
$ sudo apt update
$ sudo apt upgrade
```

7. Install the following packages:

```
$ sudo apt install build-essential vim openssh-server
```

   Note: If the keyboard layout is not correct, in the VM's OS, go to *Settings* → *Keyboard* → *Add* a new input (Portuguese) and remove the old one (English).

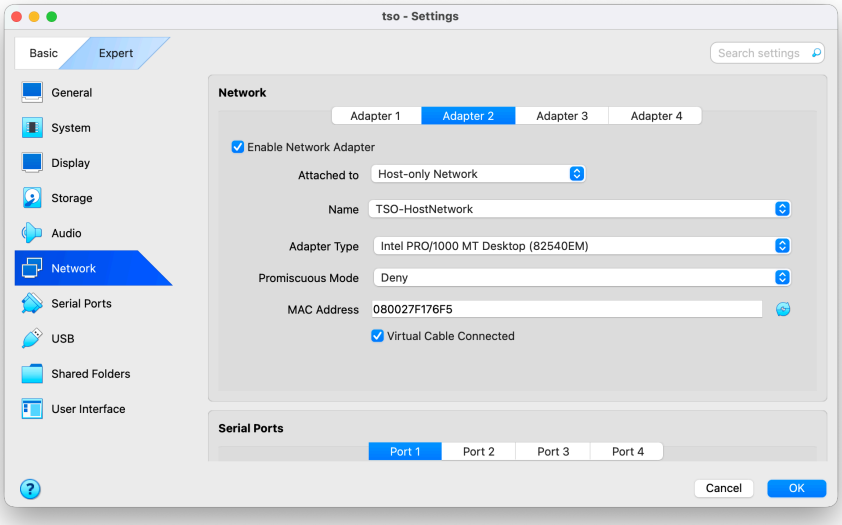
8. Install the VM Guest Additions package by selecting, in your host's OS, the menu *Devices* → *Insert Guest Additions CD Image*. In the VM's OS, open the folder containing the package (/media/<username>/VBox_GAs_7.2.4) and run the program.

9. Power off the VM.

2.2 Remote connection

2.2.1 Attach the Host-only Adapter

1. With the VM powered off, open *Settings* → *Network*, enable *Adapter 2*, set it to *Host-only Network*, and choose the host-only network you created in Section 1.



2. Click on "Ok" and start the VM.

2.2.2 Define a static IP for the VM

1. Use the VirtualBox console to connect to the VM. Open the terminal application and edit the file `50-cloud-init.yaml`, for example, using the `vim` text editor:

```
$ sudo vim /etc/netplan/50-cloud-init.yaml
```

2. Edit the file by adding below `enp0s8`, the new IP configuration for your host-only network adapter.

```
network:
  version: 2
  ethernets:
    enp0s8:
      dhcp4: true
    enp0s9:
      dhcp4: false
      addresses:
        - 192.168.57.10/24
```

3. Finally, apply the previous configurations.

```
$ sudo netplan apply
```

4. Test that the SSH service daemon (sshd) is running with:

```
$ sudo systemctl status ssh.service
```

Note: If the service is not active, run the following commands:

```
sudo ssh-keygen -A
sudo systemctl start ssh.service
```

2.2.3 Connect to the VM

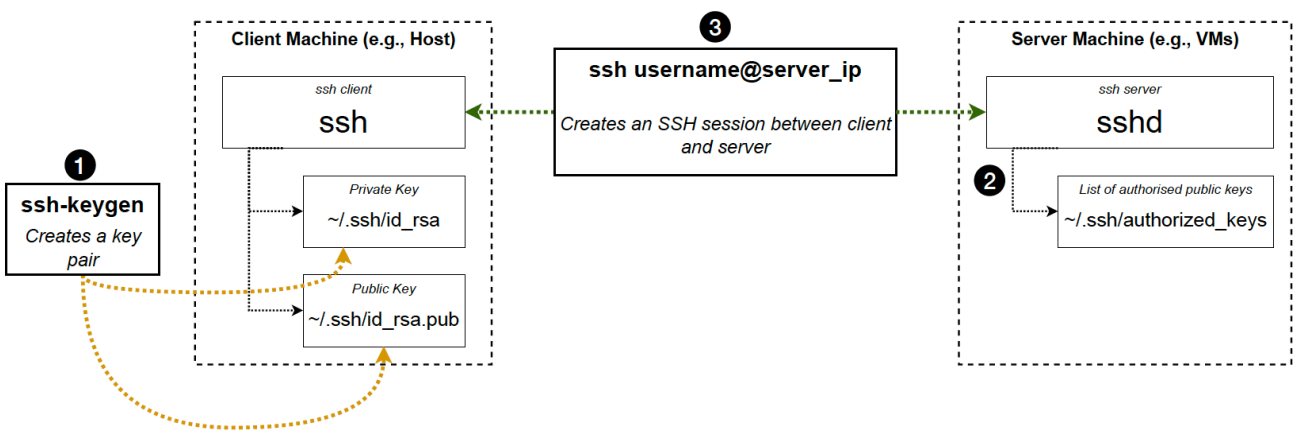


Figure 1: Flow for SSH keys generation, adding the public key to the remote server, and connecting to the server through the SSH protocol.

1. Open a terminal on your computer (host OS) and use the SSH client to connect to the VM.

```
$ ssh <USERNAME>@192.168.57.10
```

2. Create an SSH key. Instead of using the password, one can use SSH keys to authenticate at remote servers (the VM in this case). This is a stronger authentication method that uses public key cryptography to authenticate hosts and users.

   Use the following command (on your host) to create a new pair of SSH keys:

   ```
   $ ssh-keygen
   ```

   Note: The **private key** (e.g., id_rsa) is secret, known only to the user, and should remain stored safely (e.g., in the user's laptop accessing the remote server). The **public key** (e.g., id_rsa.pub) can be shared freely with any SSH server to which the user wishes to connect (e.g., VM).

3. Copy the SSH key to the VM:

   ```
   $ ssh-copy-id <USERNAME>@192.168.57.10
   ```

   Note. Alternatively, if your OS does not support the previous command, copy your public key's content and add it to the `~/.ssh/authorized_keys` file in your VM. The **authorized_keys** file specifies the SSH public keys that can be used for logging into the remote server.

4. Try connecting again with the `ssh <USERNAME>@192.168.57.10` command.

   ```
   $ ssh [-i <PATH/TO/KEY>] <USERNAME>@<IP_ADDRESS>
   ```

5. To improve security, disable password authentication and allow only public key authentication by editing the */etc/ssh/sshd_config* file (with `sudo`) and updating the following settings:

   - `#PubkeyAuthentication yes` → `PubkeyAuthentication yes` (without the #)
   - `#PasswordAuthentication yes` → `PasswordAuthentication no` (without the #)

6. Restart the SSH service in the VM.

   ```
   $ sudo systemctl restart ssh.service
   ```

   From now on, you can only connect to your VM using the configured SSH key.

2.3 Bash and SSH

Explore the following commands and check their usage and/or output:

- `cd` - move through directories;
- `man` - manual pages for commands, library functions, system calls, etc.;
- `vim` / `nano` - text editors;
- `cat` / `more` / `less` / `head` / `tail` - file content visualization;
- `mkdir` / `rmdir` - directory manipulation;
- `ls` / `pwd` / `cd` - file system navigation;
- `mv` / `cp` / `rm` - file manipulation;
- `scp` / `rsync` - file or directory transferring;
- `grep` - search for a particular pattern;
- `find` - finding libraries, directories or files;
- `sort` - ordering the file lines in txt format;

Also, explore the following bash operators:

- `>`, `<`, `>>`, `<<` - redirect program input/output to file;
- `|` - compose different commands (operator pipe).

Suggestion: Take a look at the *bashintro.pdf* file for further information and examples about all these commands and operators.

2.4 Fun

1. Install the SuperTux game and enjoy! You need to use the GUI console to play...

   ```
   $ sudo snap install supertux
   ```